

Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

Oğuz Umut Salman

Laboratoire des Sciences des Procédés et des Matériaux (LSPM)
CNRS UPR 3407 | Université Sorbonne Paris Nord

31 Mars 2026

Rapporteurs

Mme Maurine MONTAGNAT
M. Muhittin MUNGAN
M. Aurélien VATTRÉ

DR CNRS, IGE
Professeur, Univ. of Cologne
MR Onera, Univ. Paris-Saclay

Examineurs

Mme Véronique LAZARUS
M. Nicolas AUFRAY
M. Denis GRATIAS
M. Damien FAURIE
M. Ioan R. IOANESCU

Professeure, ENSTA Paris
Professeur, Sorbonne Université
DR émérite CNRS, IRCP
Professeur, USPN
Professeur, USPN

Scientific Context: Nonlinearity & Instability

The Challenge: Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

- ▶ **Phase Transitions:** Martensitic transformations.
- ▶ **Plasticity:** Dislocation-mediated flow and avalanches.
- ▶ **Fracture:** Brittle-to-ductile transitions and damage.

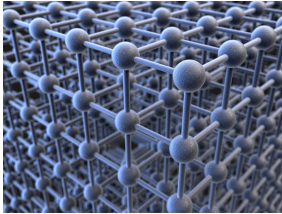
Unifying Perspective

Treating these diverse phenomena as *displacive instabilities* driven by the competition between elastic energy and lattice symmetry.

Non-linear phenomena

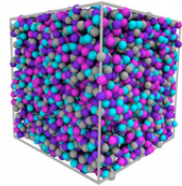
Crystal Transitions and Disorder

Crystalline



Stronger disorder
Weak disorder

Amorphous



Meta-materials

spring-chain



spring-chain system embedded into a pantograph

